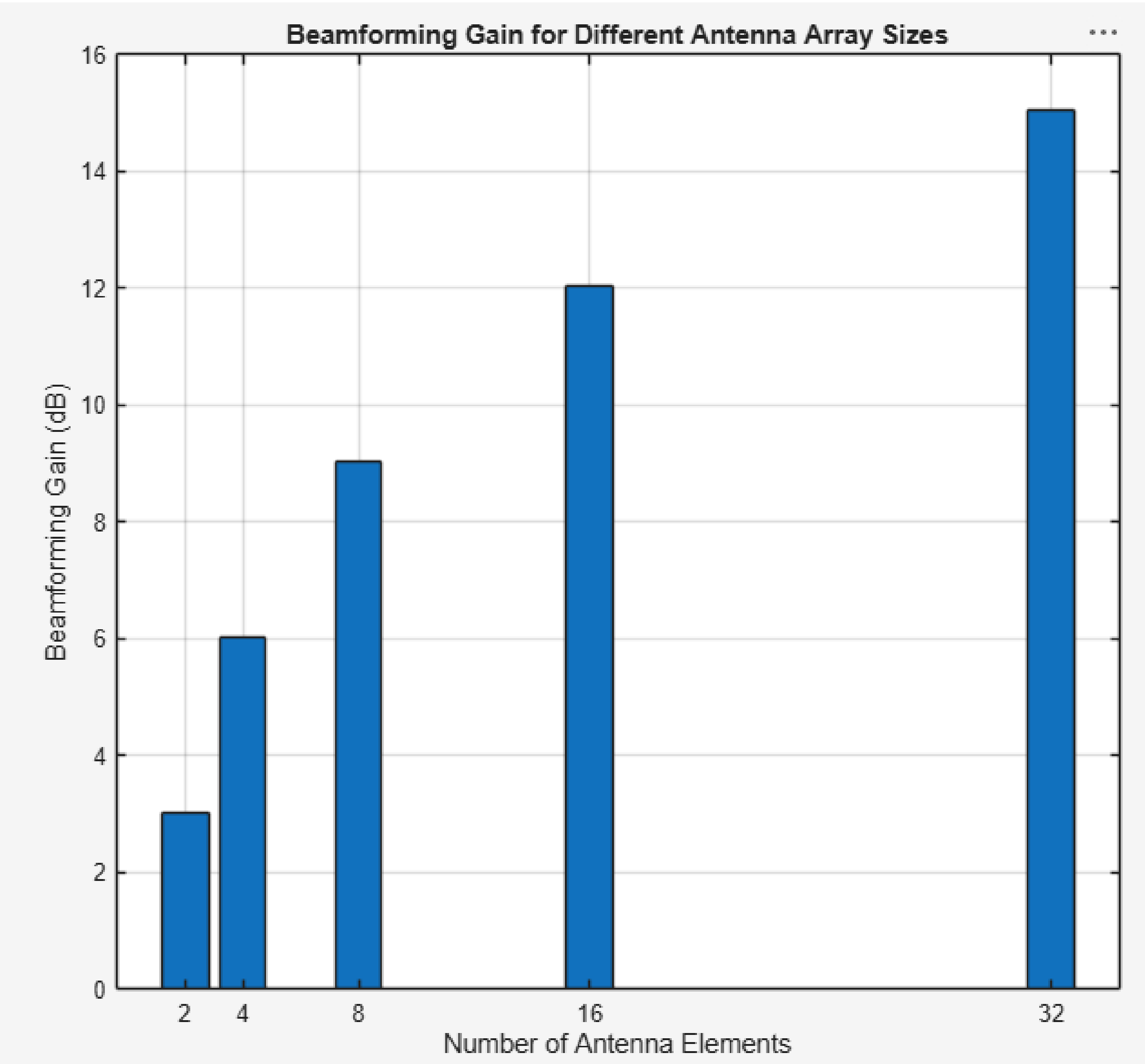
Ex.No.7: Analog Beamforming Analysis Using MATLAB

Objective 1: Matlab Code and Output

/MATLAB Drive/slotb7.m

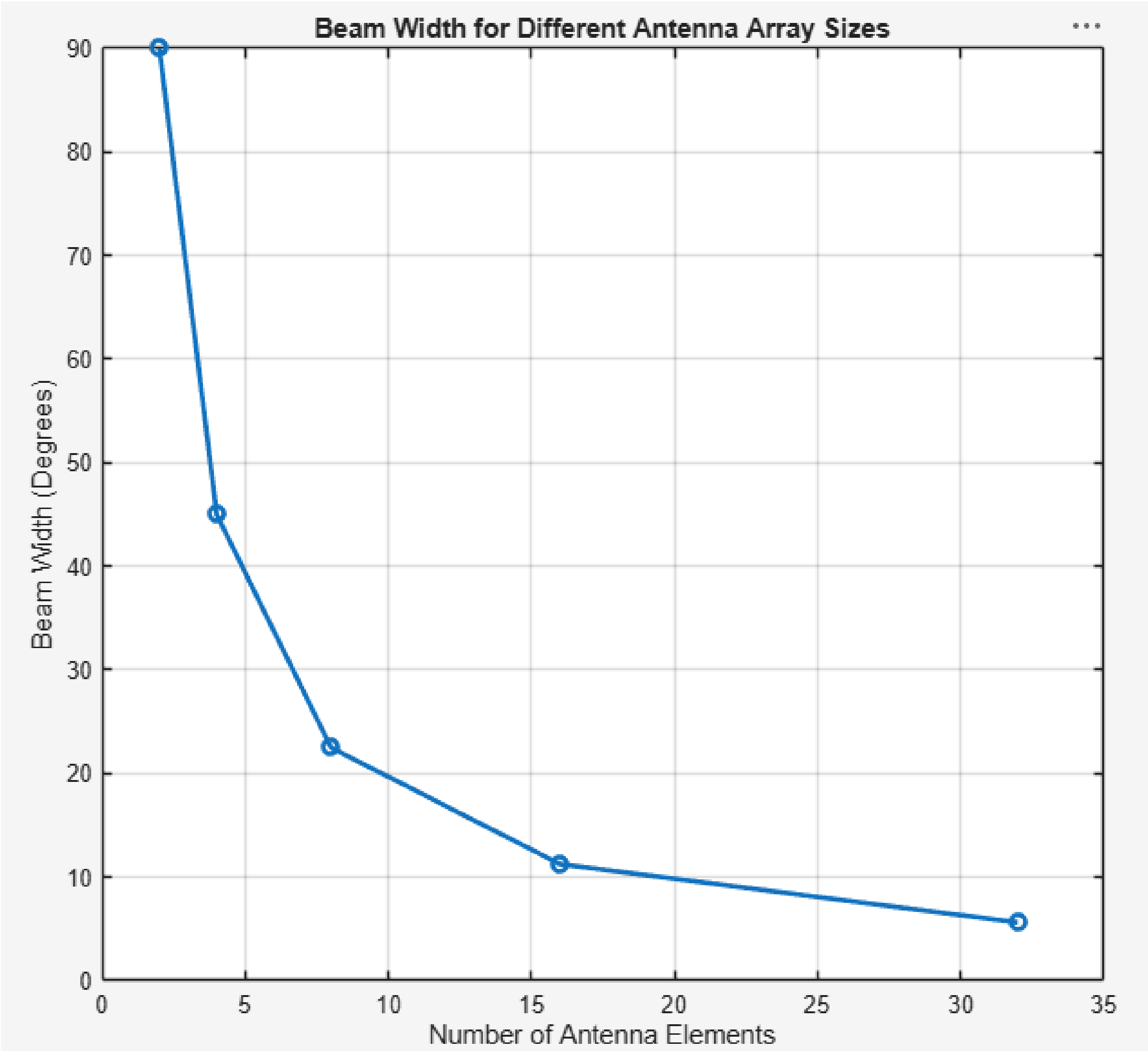
```
clc;
clear;
close all;
antennas = [2 4 8 16 32];
gain = 10*log10(antennas);
figure
bar(antennas, gain)
grid on
% Number of Antenna Elements
% Beamforming Gain (dB)
xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')

```



Objective 2: Matlab Code and Output

```
antennas = [2 4 8 16 32];
% Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625];
figure
% Beam Width (degrees)
plot(antennas, beamwidth,'o-','LineWidth',2)
grid on
xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')
```



Objective 3: Matlab Code and Output

```
theta = [-60 -30 0 30 60];
gain = [1 1 1 1 1];
figure
% Main Lobe Direction (degrees)
% Constant normalized gain
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```